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1  cd "F:\ECON 408\Practice do
   file\Assignments\Time-Series-Assignment\Assugment-45\Do
   file"
2
3  clear all
4  set more off
5  set seed 10
6
7  set obs 150
8  gen t = _n
9  tsset t
10
11 set scheme white_tableau
12 graph set window fontface "CMU Serif"
13 graph set eps fontface "CMU Serif"
14 graph set svg fontface "CMU Serif"
15
16 * Step-1: Augmented Dicky Fuller Test of 2 variables
17 *-----
   -----
18
19 * Random Walk with a drift of 0.2
20
21 gen double e = rnormal()
22 gen y_drift = sum(0.2 + e)
23
24 tsline y_drift, ///
25         lcolor(ebblue%70) lwidth(medthick) ///
26         title("{bf:Time Series Plot}", size(large)) ///
27         subtitle("Random Walk with a drift of 0.2") ///
28         xtitle("Time Period", size(medlarge)) ///
29         ytitle("Values", size(medlarge)) ///
30         ylabel(, labsize(medium) nogrid) ///
31         xlabel(, labsize(medium) nogrid) ///
32         graphregion(color(white)) xsize(8) ysize(4) ///
33         name(tsplot_hw4_1, replace)
34
35 graph export tsplot_hw4_1.png, name(tsplot_hw4_1) replace
36
37 varsoc y_drift, maxlag(12)
38 dfuller y_drift, trend lags(1)
39

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40
41 * AR(1), rho = 0.96, deterministic trend of 0.2t.
42
43 gen double u = e in 1
44 replace u = 0.96*L.u + e in 2/L
45 gen double y_trend = 0.2*t + u
46
47
48 tsline y_trend, ///
49         lcolor(ebblue%70) lwidth(medthick) ///
50         title("{bf:Time Series Plot}", size(large)) ///
51         subtitle("AR(1) Deterministic trend of 0.2t with
rho = 0.96 ") ///
52         xtitle("Time Period", size(medlarge)) ///
53         ytitle("Values", size(medlarge)) ///
54         ylabel(, labsize(medium) nogrid) ///
55         xlabel(, labsize(medium) nogrid) ///
56         graphregion(color(white)) xsize(8) ysize(4) ///
57         name(tsplot_hw4_2, replace)
58
59 graph export tsplot_hw4_2.png, name(tsplot_hw4_2) replace
60
61
62 varsoc y_trend, maxlag(12)
63 dfuller y_trend, trend lags(1)
64
65
66 *Step-2: Generating and Simulating Mone-Carlo Simulation
67 *-----
68
69
70 capture program drop sim_power
71 program sim_power, rclass
72
73     syntax [, obs(integer 150)]
74
75     quietly {
76
77         drop _all
78         set obs `obs'
79         gen t = _n

```

```

80         tsset t
81
82         * Random Walk with a drift of 0.2
83
84         gen double e = rnormal()
85         gen y_drift = sum(0.2 + e)
86         dfuller y_drift, trend lags(1)
87
88         return scalar t_drift = r(Zt)
89
90         * AR(1), rho = 0.96, deterministic trend of
91         0.2t.
92
93         gen double u = e in 1
94         replace u = 0.96*L.u + e in 2/L
95         gen double y_trend = 0.2*t + u
96         dfuller y_trend, trend lags(1)
97
98         return scalar t_trend = r(Zt)
99     }
100 end
101
102 simulate t_drift = r(t_drift) t_trend = r(t_trend), reps(
103 1000) seed(10): sim_power, obs(150)
104
105 * STEP 3: Analyze and Visualize the T-statistic
106 *-----
107 --
108
109 gen reject_drift = (t_drift < -3.44)
110 gen reject_trend = (t_trend < -3.44)
111
112 display "--- SIMULATION RESULTS (PERCENTAGE OF
113 REJECTIONS) ---"
114
115 sum reject_drift reject_trend
116
117 twoway ///
118     (kdensity t_drift, lcolor(pink) lwidth(medthick)) ///

```

```
118      (kdensity t_trend, lcolor(ebblue%70) lwidth(medthick
119      )), ///
120      title("Augmented Dickey-Fuller Test Power") ///
121      subtitle("Overlap of t-statistic distributions
122      (T=150)") ///
123      xtitle("t-statistic") ///
124      ytitle("Density") ///
125      xlabel(,nogrid) ylabel(,nogrid) ///
126      xline(-3.44, lcolor(black) lpattern(dash) lwidth(
127      medthick)) ///
128      legend(order(1 "Null: rho=1.00" 2 "Alt: rho=0.96") ///
129      col(1) pos(2) ring(0)) name(hw4_kdensity,
130      replace)
131
132      graph export hw4_kdensity.png, name(hw4_kdensity) replace
133
```